

Sun, Chemicals & the Nail

UV lamp exposure, chemical ventilation, and the single habit that protects more than any product you will ever sell

Learning Objectives

- Explain what UV nail lamps emit, and the current evidence on their skin-safety profile
- Recommend a simple protective habit for clients receiving regular gel manicures
- Explain the ventilation and inhalation risks in a working nail station
- Apply correct chemical-safety practice for acetone, primers, and acrylic dust

Two different exposures, one working station

A busy nail station carries two separate categories of chemical and radiation exposure — light-based, from the UV/LED lamps curing gel introduced in Module Four, and chemical, from the solvents, primers, and dust generated during filing and removal. Both deserve calm, evidence-based attention, not panic and not dismissal.

UV nail lamps: what the evidence actually shows

Gel-curing lamps emit primarily UVA radiation, the same wavelength responsible for skin ageing and, with significant cumulative exposure, increased skin cancer risk, as taught in Course I. Research directly measuring nail-lamp exposure has generally found the dose per session to be low compared to the reference points used for photoageing and skin cancer research, and current expert guidance does not treat occasional gel manicures as a major individual risk. That said, "low per session" is not "zero cumulative," particularly for clients receiving gel manicures every two to three weeks for years. The professionally responsible response is not to abandon gel services — it is to reduce the exposure simply and reliably.

◆ Figure 7.1 · Two Exposures, One Station



Two distinct exposure categories, each with its own simple, reliable mitigation.

The single habit that matters more than any product

Just as this Atelier has taught in every course before this one, the most effective protection is a simple habit applied consistently: **apply a broad-spectrum sunscreen to the hands, or offer fingerless UV-protective gloves, before every gel-curing session under the lamp.** This single step meaningfully reduces UVA exposure to the surrounding skin without in any way interfering with the gel curing on the nail plate itself. Make it a standard, offered step in every gel service, not an optional upsell.

Ventilation and chemical safety

- **Acetone and other solvents.** Work in a well-ventilated space, ideally with a dedicated extraction fan at each station, and keep solvent containers capped between uses to limit vapour buildup.
- **Acrylic and gel primers.** Some contain strong acids or bonding agents that can irritate skin on contact; apply precisely and avoid skin contact rather than a liberal, imprecise application.
- **Filing dust.** Acrylic and gel dust generated during removal and shaping can irritate the respiratory tract with repeated occupational exposure; a dust-collecting table vent or, at minimum, a well-ventilated station and an occasional mask during heavy filing sessions meaningfully reduces a practitioner's own cumulative exposure across a working career.

Key Terms

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[object Object]

1

[object Object]

2

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Practical Activity · The Protection Habit

During your next three gel services (practice or real), apply sunscreen or offer UV-protective gloves before every lamp cure without exception. Note whether clients had heard of this practice before, and how they responded to it being offered.